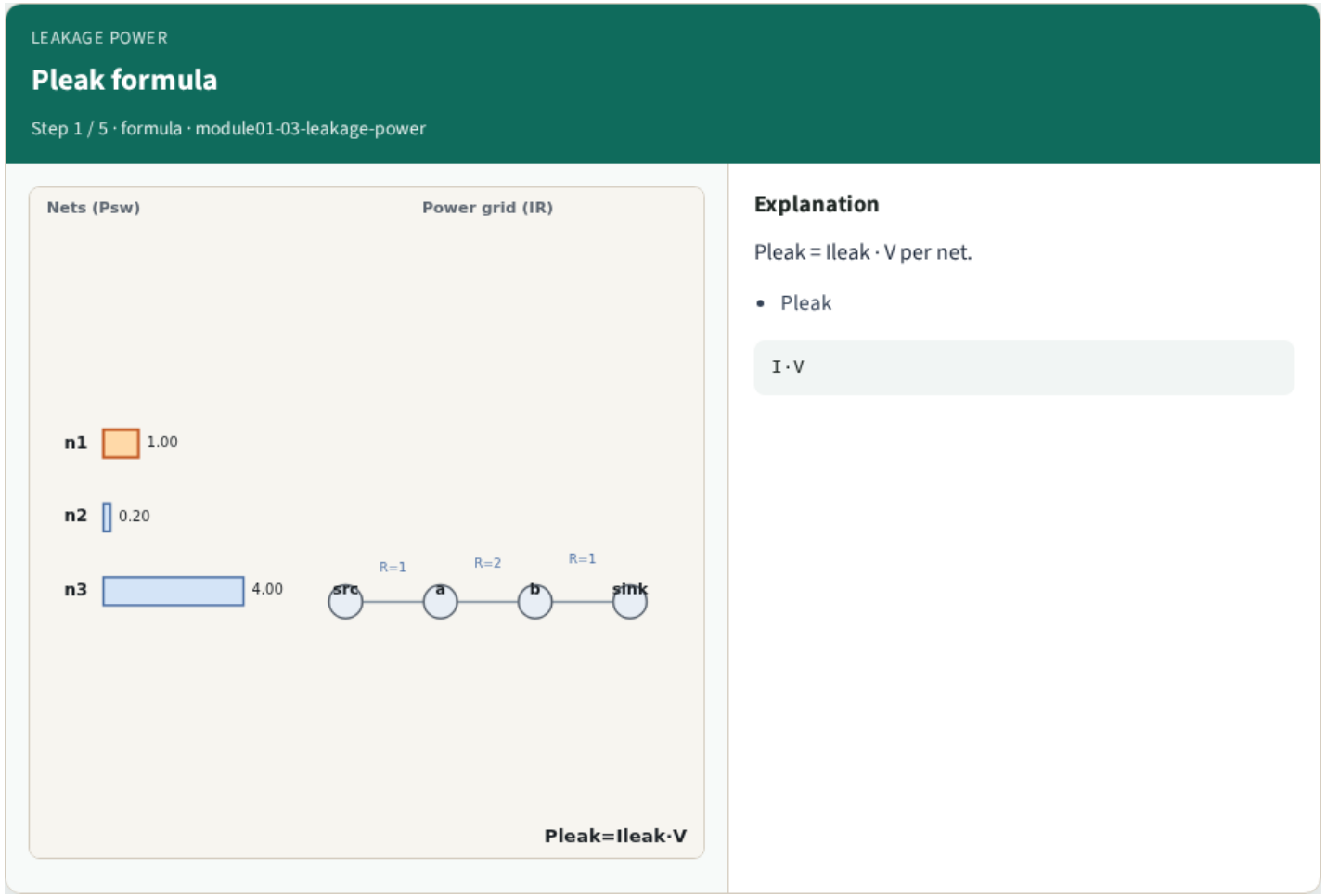
Why leakage

Static leakage current times supply voltage gives Pleak

The idea

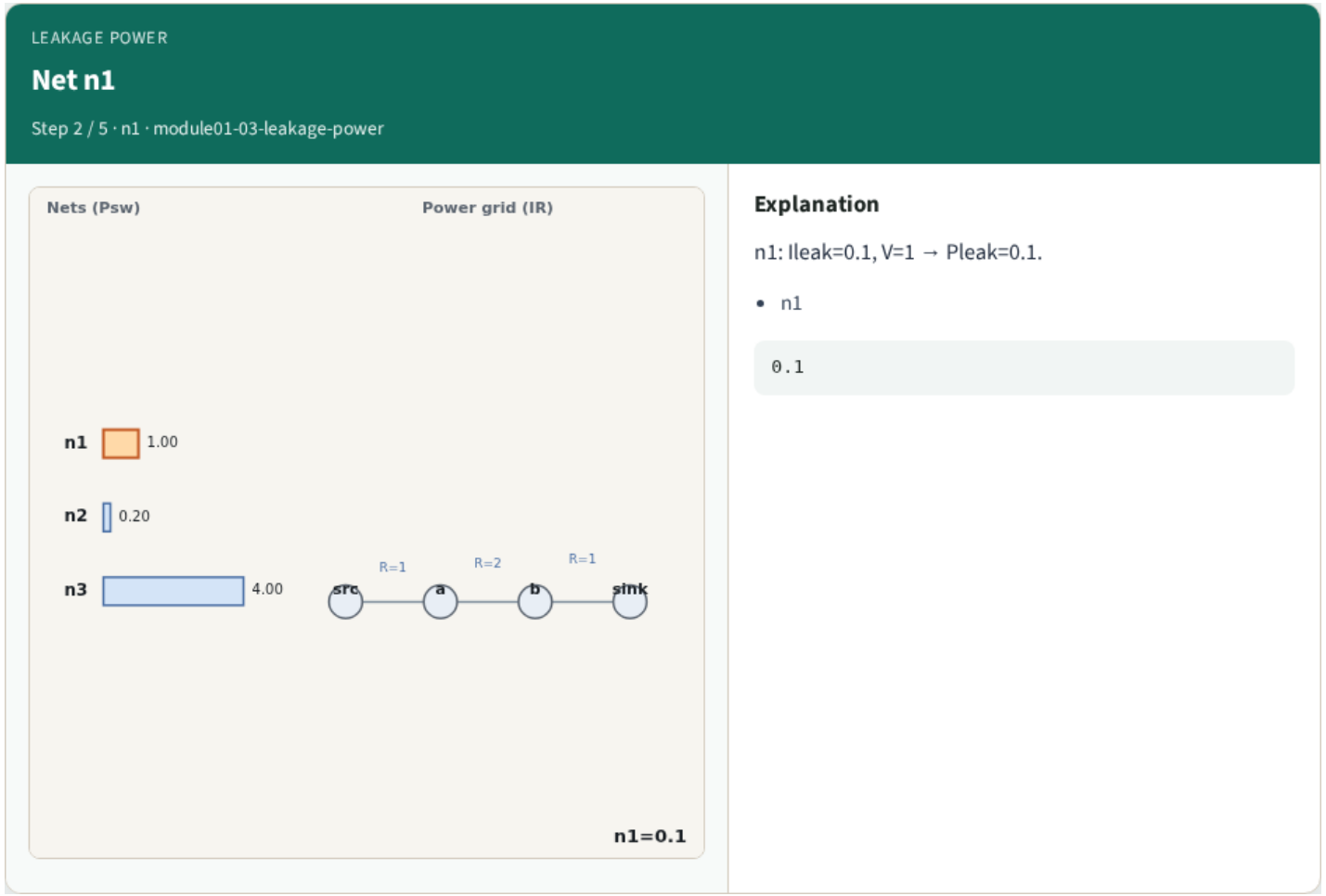
- GOLDEN: plk_n1=0.1

Pleak formula



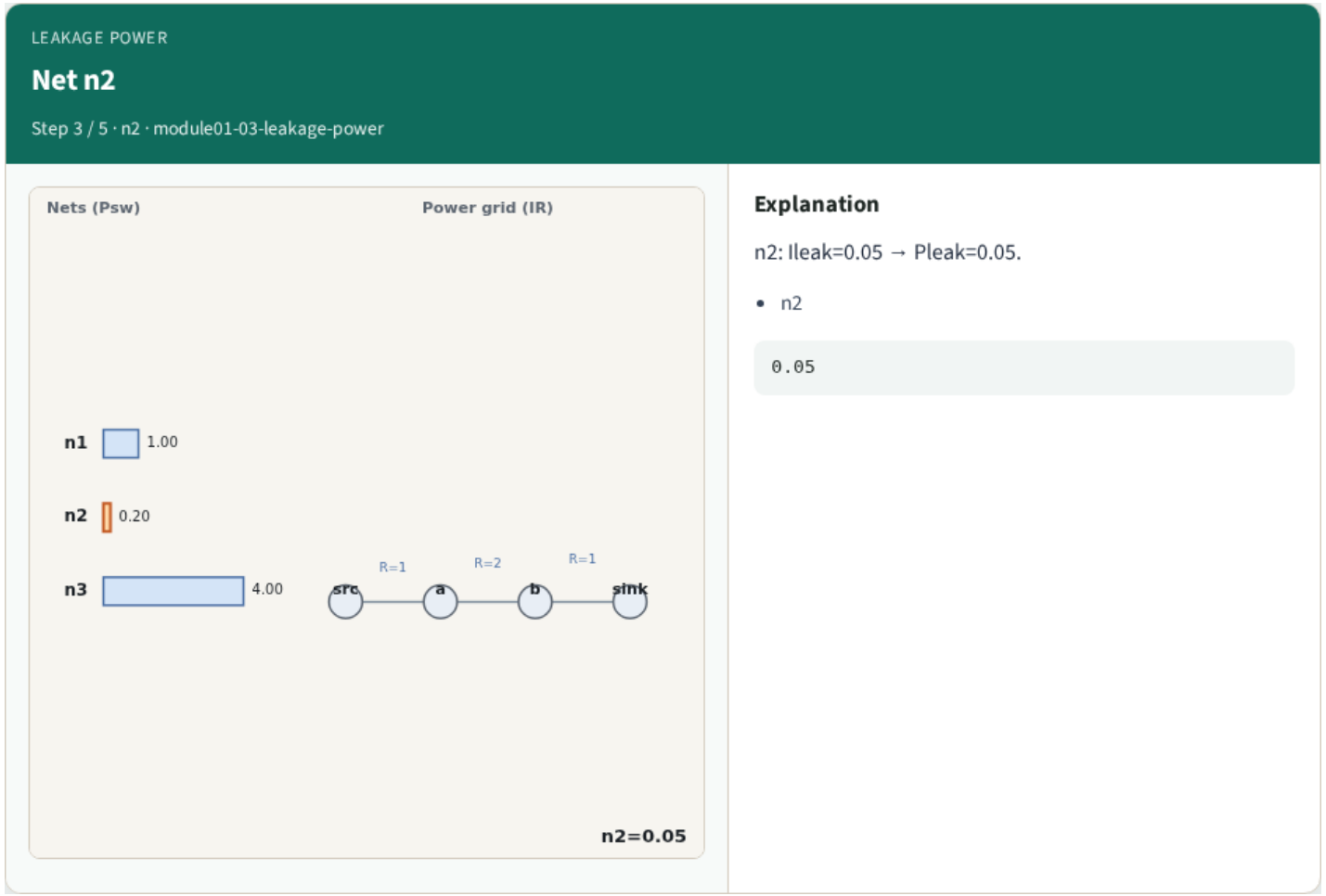
Pleak formula

Net n1


LEAKAGE POWER
Net n1
Step 2 / 5 · n1 · module01-03-leakage-power
Nets (Psw)
Power grid (IR)
n1
1.00
n2
0.20
n3
4.00
R=1
R=2
R=1
src
a
b
sink
n1=0.1
Explanation
n1: Ileak=0.1, V=1 → Pleak=0.1.
n1
0.1

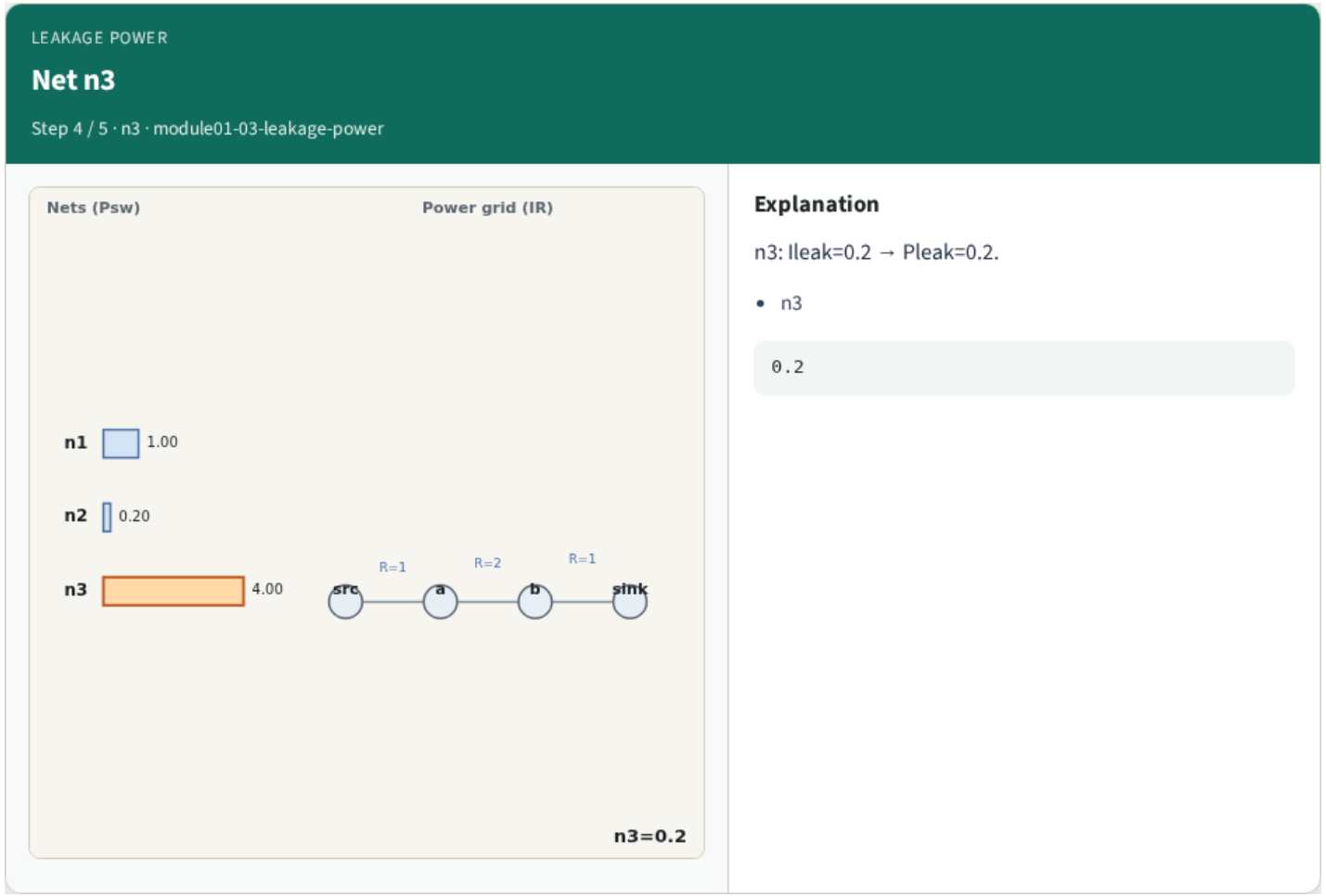
Net n1

Net n2


LEAKAGE POWER
Net n2
Step 3 / 5 · n2 · module01-03-leakage-power
Nets (Psw)
Power grid (IR)
n1
1.00
n2
0.20
n3
4.00
R=1
R=2
R=1
src
a
b
sink
n2=0.05
Explanation
n2: Ileak=0.05 → Pleak=0.05.
n2
0.05

Net n2

Net n3


LEAKAGE POWER
Net n3
Step 4 / 5 · n3 · module01-03-leakage-power
Nets (Psw)
Power grid (IR)
n1
1.00
n2
0.20
n3
4.00
R=1
R=2
R=1
src
a
b
sink
n3=0.2
Explanation
n3: Ileak=0.2 → Pleak=0.2.
n3
0.2

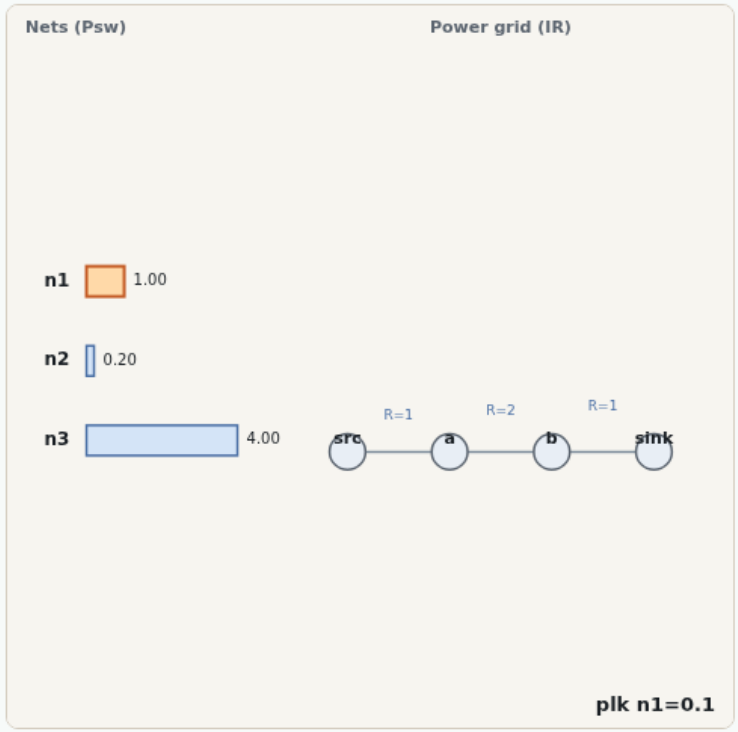
Net n3

Golden

LEAKAGE POWER

Golden

Step 5 / 5 · golden · module01-03-leakage-power



Explanation

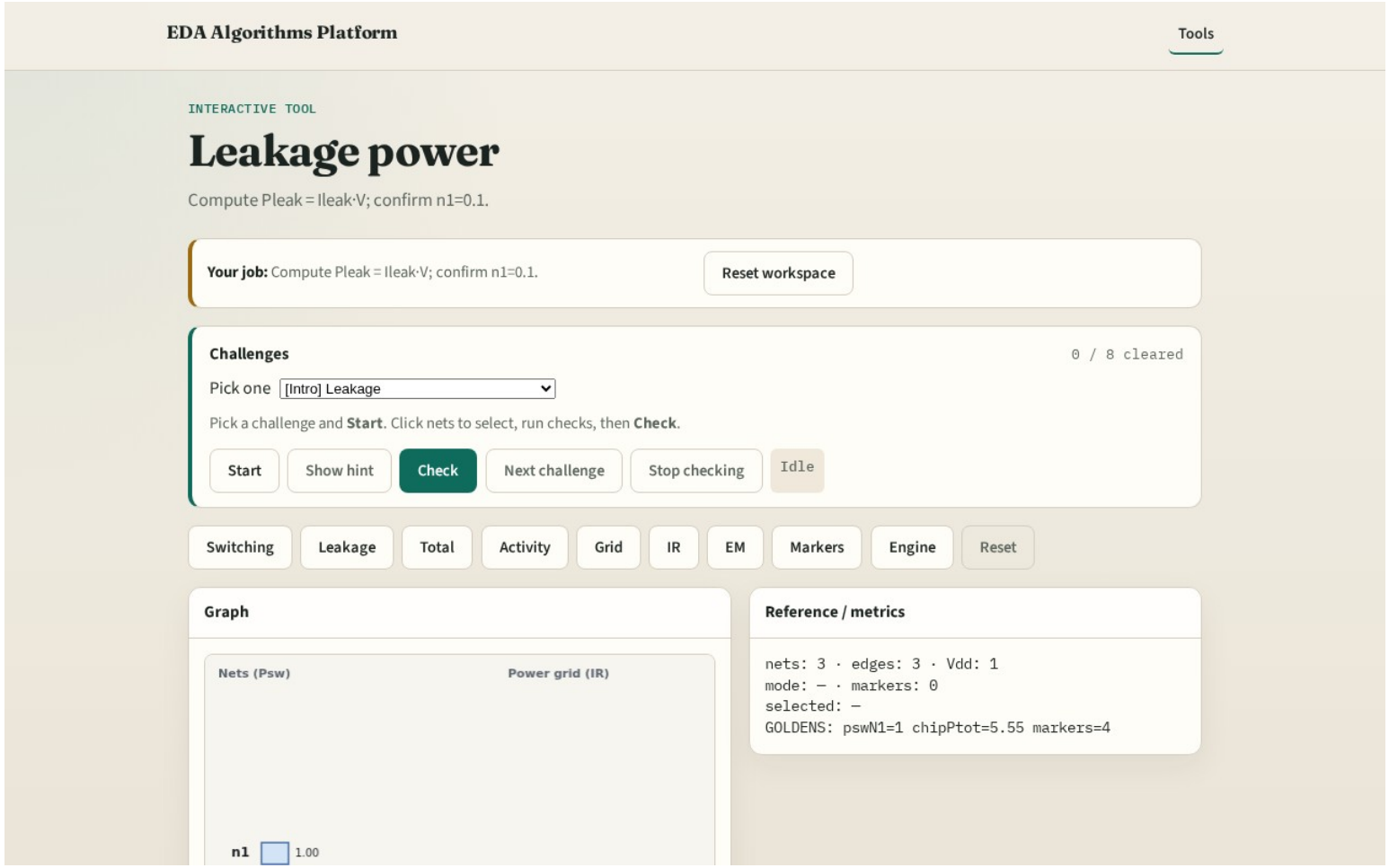
GOLDEN: plk_n1=0.1; chip_plk=0.35.

- GOLDEN

```
0.1/0.35
```

Golden

Browser lab track



Browser lab starter

Implement track

- Call leakage_power(); assert by_net['n1']==0.1

Pitfalls

- Mixing Ileak with switching alpha
- Wrong voltage

Your turn

- Next: total power